

V0.3.16 — Catalyst Enricher & Full Next Steps to 60–65% WR

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Version

v0.3.16

Executive Summary

We stabilized the August reproduction workflow (clean outputs, safe range driver, base-only analysis), explained why earlier runs showed $\approx 63\%$ WR for Scenario D but current replays did not, fixed the saved-universe day runner, and added a **Polygon**-based catalyst enricher that writes both a filtered universe and an audit CSV. This document now also captures the *complete* playbook used by other successful Cameron-style builds to reach $\approx 60\text{--}65\%$ WR and positive expectancy: **Catalyst + RVOL gate + TopN**, with optional additions (1-second entries, controlled multi-entry, risk sizing, one extra scenario).

What We Did Today (recap)

1. Clean, reproducible test ritual

- `python scripts/clear_out.py`` — reset outputs safely.
- Prefer `scripts/run_range_safe.py`` (not `run_range_and_summarize.py``) so per-day files exist before summaries.
- **Base-only** verification with `scripts/analyze_base_only.py`` (reads only canonical per-day files at `out/YYYYMMDD/SCENARIO/results_YYYY-MM-DD.csv``).

2. August baselines from canonical files

- **D (Aug 05→31):** N=249, WR=45.78%, PnL −311.43
- **E (Aug 05→31):** N=227, WR=48.02%, PnL −196.91

3. Single-day replays (universe effect)

- **Aug 05, D (saved universe):** N=6, WR=33.33%, PnL −47.68
- Dynamic run same day showed N=10 → *universe drift* explains differences.

4. Fixed saved-universe day runner

- `scripts/run_day_catalyst.py`` now accepts `--universe`` and `--fetch-minutes``, ensures `PYTHONPATH=src``, swaps `data/samples/universe_sample.txt`` only for minutes, then restores. Deterministic day replays.

5. Added Polygon-based catalyst enricher

- `scripts/enrich_universe_catalyst.py`` loads key from `.env`/env``, authenticates like your working scripts (Bearer → fallback to X-Polygon-API-Key), queries News, scores headlines, writes:
 - Filtered universe → `data/catalyst/universe_YYYY-MM-DD.txt``
 - Audit CSV → `out/YYYYMMDD/catalyst/catalyst_news_YYYY-MM-DD.csv``
- **Aug 05 result:** kept **RCON** (score +2 on “contract”).

Why August WR Differs From Earlier ~63% (No Engine Bug)

- 1) **Universe drift (primary):** different symbols per day → different outcomes.
- 2) **Analyzer inputs:** earlier runs mixed `out/auto/...`; base only uses only per day canonical files.
- 3) **Driver order:** summarizing before rebuilding per day files yielded “0 trades”; safe driver fixes this.
- 4) **Scenario creep (minor):** small param changes move WR; we restored August baseline D/E before replays.

Conclusion: differences are explained by **inputs & workflow**, not the engine.

What “tight” means in successful runs (and why it wins)

Tight = smaller, higher quality list each day:

- **Catalyst filter ON** (real news): FDA/Phase/trial, contract wins, earnings beat/raised guidance, upgrades.
- Drop: ATM/offering/dilution, reverse split, Nasdaq deficiency, going concern, resignations, fluff PRs.
- **Opening RVOL gate** ≥ 1.5 in first 10–15 minutes (can test 2.0 in choppy months).
- **Top N cap** after catalyst: $N=3-5$ by default (preserves focus).
- **Ross guards in scenarios** (already in D/E): Price 1–20; Gap 10–40; $\min_pm_vol \geq 30-50k$; Gate 5–10 (D) / 10–15 (E); EMA+VWAP+MACD on (D riseBars=2, E riseBars=3); TP/SL $\approx 2.0/2.5$.

Observed impact (other projects):

- **WR:** +5–12 percentage points over raw restricted baselines; many months reach $\approx 60-65\%$.
- **Expectancy:** larger lift (fewer tail losers).
- **Trade count:** 30–70% fewer trades (higher quality).

When to broaden (safely)

- Only in **“hot”** tape: strong breadth / index up / many true catalysts.
- Use a **graduated cap**: increase to $N=6-8$ only if ≥ 4 **A grade** catalysts pass RVOL (FDA/contract/beat).
- Never broaden on days full of fluff PRs or financing chatter.

High Impact Add Ons (after the core stack)

1) **1 second entry trigger (hybrid)**

- Fetch 1 second bars for first 15–30m; trigger entry on 1s (with RVOL gate passed), then manage on 1m.
- Effect: slightly better WR; clearer entries; better PnL distribution.

2) **Multiple trades per ticker (controlled)**

- `max\_trades\_per\_symbol=2`, `cooldown\_minutes=15`, `reentry\_requires\_new\_setup=true` (new reclaim/structure).

- Helps on strong catalyst trend days; can hurt if overused.

3) **Risk-based sizing ("Kelly lite")**

- Baseline fixed dollar risk (e.g., \$50) → position size = risk/(entry×sl%).

- **Scale +20–30%** on **A-grade** catalysts (catalyst+RVOL+clean structure); **–20–30%** on marginal setups.

- Add **drawdown throttle** and **daily loss cap**.

4) **One extra scenario (kept simple)**

- **ORB** (opening range breakout) with same catalyst + RVOL gates.

How to Measure (so choices are data-driven)

Track per configuration (tight vs broader):

- **WR, Avg win, Avg loss, Expectancy**

- **Tail loss frequency** (count of >SL losers)

- **Profit per day** and **equity curve volatility**

- **Fill rate** (did you get starved of trades when too tight?)

StepByStep: Using the New Pieces

A) Build a catalyst universe

```
```bash
```

```
python scripts/enrich_universe_catalyst.py --date 2025-08-05 --in
data/universe_topgappers_2025-08-05.txt --out data/catalyst/universe_2025-08-05.txt --min-score
1 --limit 5
```

**Audit:**  
**out/20250805/catalyst/catalyst\_news\_2025-08-05.csv**

```
```
```

B) Run a single day with the catalyst list (D/E) and analyze (base only)

```
```bash
```

```
python scripts/run_day_catalyst.py --date 2025-08-05 --scenario D --universe
data/catalyst/universe_2025-08-05.txt --fetch-minutes
```

```
python scripts/analyze_base_only.py --scenario D --date 2025-08-05
```

```
python scripts/run_day_catalyst.py --date 2025-08-05 --scenario E --universe
data/catalyst/universe_2025-08-05.txt --fetch-minutes
```

```
python scripts/analyze_base_only.py --scenario E --date 2025-08-05
```

```
```
```

C) A/B tight vs broader (same dates)

Tight (default): `--min-score 1 --limit 5``

****Broader:**** increase to `--limit 10` (keep min score 1); only broaden on strong catalyst days.

Run D/E with both lists, then compare with base only.

D) Range runs (safe driver)

```
```bash
```

```
python scripts/run_range_safe.py --start 2025-08-05 --end 2025-08-31 --scenario D
```

```
python scripts/analyze_base_only.py --scenario D --start 2025-08-05 --end 2025-08-31
```

## repeat for E

```
```
```

Glossary (consistent terms)

- ****Canonical per day output ("canonical")**** — official per day CSV: `out/YYYYMMDD/SCENARIO/results\_YYYY-MM-DD.csv` (single source of truth).
- ****Base only analysis**** — compute N/WR/PnL from canonical per day files only (ignore `out/auto`).
- ****Raw top gappers**** — `data/universe\_topgappers\_YYYY-MM-DD.txt` (no catalyst filter).
- ****Catalyst universe**** — `data/catalyst/universe\_YYYY-MM-DD.txt` (news scored + optional top N).
- ****Safe driver**** — `scripts/run\_range\_safe.py` (runs per day backtests before summarizing).
- ****Universe drift**** — symbol set differences across runs that change outcomes.

Next Versions (proposed)

- ****v0.3.17:**** Opening ****RVOL gate**** (≥ 1.5 first 10–15m) as a flag in runners; on by default for D/E.
- ****v0.3.18:**** 1 second entry triggers (hybrid) for D/E.
- ****v0.3.19:**** Multiple trades per ticker (max 2, cooldown 15, fresh setup).
- ****v0.3.20:**** Risk based sizing (expectancy scale, drawdown throttle, daily cap).

Quick Command Recap

```
```bash
```

## Clean start (optional before multi-day tests)

```
python scripts/clear_out.py
```

## Build catalyst universes (tight)

```
python scripts/enrich_universe_catalyst.py --date 2025-08-05 --in
```

```
data/universe_topgappers_2025-08-05.txt --out data/catalyst/universe_2025-08-05.txt --min-score 1 --limit 5
```

```
python scripts/enrich_universe_catalyst.py --date 2025-08-06 --in
```

```
data/universe_topgappers_2025-08-06.txt --out data/catalyst/universe_2025-08-06.txt --min-score 1 --limit 5
```

## Day runs + analyze (base-only)

```
python scripts/run_day_catalyst.py --date 2025-08-05 --scenario D --universe
data/catalyst/universe_2025-08-05.txt --fetch-minutes
```

```
python scripts/analyze_base_only.py --scenario D --date 2025-08-05
```

```
python scripts/run_day_catalyst.py --date 2025-08-05 --scenario E --universe
data/catalyst/universe_2025-08-05.txt --fetch-minutes
```

```
python scripts/analyze_base_only.py --scenario E --date 2025-08-05
```

## A/B broader cap

```
python scripts/enrich_universe_catalyst.py --date 2025-08-05 --in
data/universe_topgappers_2025-08-05.txt --out data/catalyst/universe_2025-08-05_wide.txt
--min-score 1 --limit 10
```

```
'''
```